

Supplementary Materials

The SymC Neurostability Principle: Governing Physics of Instability Architectures in Alzheimer’s and Dementia

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November 2025

S1: EEG-Based $\chi(t)$ Estimation Methodology

The damping ratio can be estimated from EEG using the Hilbert transform:

1. **Analytic Signal Construction:** Construct $z(t) = x(t) + i\mathcal{H}\{x(t)\} = A(t)e^{i\phi(t)}$, where $\mathcal{H}$ denotes the Hilbert transform, $A(t)$ is the instantaneous envelope, and $\phi(t)$ is instantaneous phase.
2. **Instantaneous Frequency Extraction:** $\omega(t) = d\phi/dt$, extracted from the dominant alpha rhythm (8 to 13 Hz).
3. **Damping Rate Estimation:** $\gamma(t) = -2(dA/dt)/A(t)$, estimated from envelope decay within sliding windows.
4. **SymC Ratio Computation:** $\chi(t) = \gamma(t)/(2|\omega(t)|)$.

Validation studies indicate reliable χ estimation requires $\text{SNR} \geq 5$ and time windows capturing at least 3 to 5 oscillation cycles.

S1.1: Theta/Alpha Ratio as Clinical Proxy

For clinical screening where Hilbert analysis may be impractical, the Theta/Alpha power Ratio (TAR) serves as a proxy for χ . Higher TAR indicates dominance of slower rhythms (reduced effective ω) and correlates with higher χ :

- Healthy elderly: $\text{TAR} < 0.7 \rightarrow \chi \approx 0.85$ to 0.95
- MCI: $\text{TAR} 0.9$ to $1.5 \rightarrow \chi \approx 0.65$ to 1.10 (U-curve)
- AD: $\text{TAR} > 2.0 \rightarrow \chi > 1.2$

S1.2: DLB-Specific: Dominant Frequency Variability

For DLB differential diagnosis, compute DFV from sequential 10-second epochs:

$$\text{DFV} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (f_{\alpha,i} - \bar{f}_{\alpha})^2} \quad (1)$$

Thresholds:

- Healthy/AD: $\text{DFV} < 0.8$ Hz
- DLB: $\text{DFV} > 1.2$ Hz

S2: Phase 2 Trial Protocol

S2.1: Study Design

Randomized, double-blind, controlled Phase 2 trial with $N \approx 300$ participants and 18-month follow-up. Stratification by baseline SymC instability, APOE $\epsilon 4$ status, dementia subtype, and site.

S2.2: Endpoint Structure

Primary Clinical Endpoints:

- Change in CDR-SB (Clinical Dementia Rating Sum of Boxes)
- Change in ADAS-Cog13

Secondary SymC Endpoints:

- *Dynamical*: Normalization of EEG-derived $\chi(t)$ into the 0.8 to 1.0 window
- *Structural*: Preservation of hippocampal stiffness ($G' > 2.5$ kPa) and damping ($\xi < 0.25$)
- *Coherence*: Restoration of 40 Hz (gamma) coherence in target regions
- *Variability*: Reduction of DFV in DLB cohort

Subtype-Specific Endpoints:

- AD: G' preservation, spectral cascade reversal
- DLB: DFV reduction, fluctuation frequency
- VaD: BP variability reduction, Fazekas stability
- FTD: Frontal-temporal coherence, NPI-disinhibition

Concordance Endpoints:

- Correlation of SymC indices with plasma p-tau217, A β 42/40, NfL
- Cross-validation of MRE-derived ξ and EEG-derived ζ

S2.3: Mediation Analysis

The statistical plan includes mediation analysis testing whether clinical improvement is mediated by SymC index changes:

$$\text{Therapy} \rightarrow \Delta \text{ SymC Index} \rightarrow \Delta \text{ Clinical Outcome}$$

If the indirect pathway (through SymC) is significant while the direct pathway is attenuated, this validates the framework’s mechanistic claims: therapies work *because* they restore χ toward the adaptive window.

S2.4: Subtype Stratification

Trial design includes pre-randomization stratification by failure mode:

- **Type 1 (Stiffness Collapse)**: AD, late-stage FTD
- **Type 2 (Controller Failure)**: DLB
- **Type 3 (Impedance Mismatch)**: VaD, mixed AD+VaD
- **Type 4 (Topology Failure)**: bvFTD, PPA

This enables detection of subtype-specific treatment effects masked in pooled analyses.

S2.5: Regulatory Pathway

The trial design supports regulatory pathways for SymC indices as Software as Medical Device (SaMD), enabling:

- Comparison across intervention modalities (pharmacological, neuromodulation, behavioral)
- Mechanistic readouts independent of specific molecular targets
- Adaptive trial designs with $\chi(t)$ -guided dose titration
- Differential diagnosis algorithms as companion diagnostics

S3: Detailed Falsification Criteria

The framework is falsified if any of the following are observed:

S3.1: AD-Specific Falsification

- Patients convert to AD without any preceding χ deviation from the adaptive window
- The U-curve trajectory is absent (χ monotonically increases or decreases throughout disease)
- The spectral cascade does not follow the predicted sequence ($\gamma \rightarrow \alpha \rightarrow \theta$)
- MRE-derived ξ and EEG-derived ζ show no concordance when expressed as χ
- ξ does not precede G' collapse (i.e., stiffness falls before damping rises)
- 40Hz entrainment efficacy shows no correlation with tissue stiffness
- Clinical improvement occurs without normalization of SymC indices

S3.2: DLB-Specific Falsification

- DLB patients show stiffness collapse equivalent to AD on MRE
- DFV does not distinguish DLB from AD at prodromal stages
- Cognitive fluctuations show no correlation with χ variance
- RBD+ cohorts show early stiffness changes rather than preserved G'

S3.3: VaD-Specific Falsification

- VaD progression is smooth rather than stepwise
- Blood pressure variability shows no correlation with cognitive outcomes
- Impedance mismatch ($|R|$) does not predict lesion location
- Focal EEG patterns indistinguishable from AD diffuse slowing

S3.4: FTD-Specific Falsification

- FTD patients show diffuse EEG slowing before behavioral symptoms
- Frontal-temporal coherence decline does not precede power changes
- Kuramoto order parameter (r) remains stable in early FTD
- MRE shows global rather than focal softening

S3.5: Cross-Diagnostic Falsification

- The four failure modes cannot be distinguished by the proposed indices
- Phase diagram clustering (G' vs. χ variability) fails to separate subtypes
- Therapeutic response is independent of baseline failure mode
- Multi-modal index does not outperform single-modality biomarkers

S4: Supplementary Figures

The Viscoelastic Cascading Failure Model (VCFM)

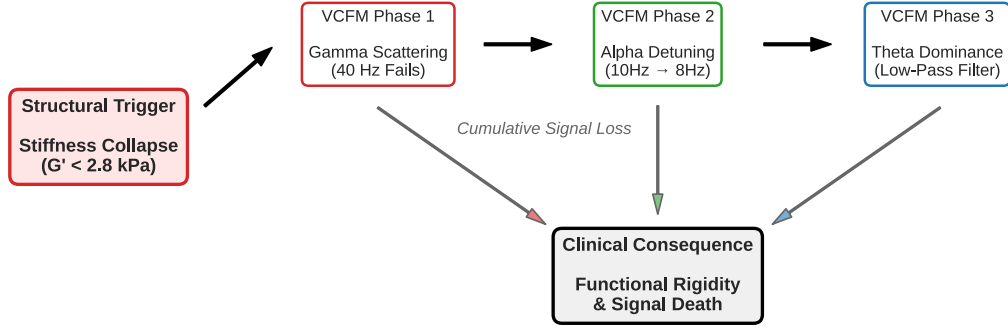


Figure 1: The Viscoelastic Cascading Failure Model (VCFM) Flowchart. Logical flow of the VCFM mechanism. The Structural Trigger (Stiffness Collapse, $G' < 2.8$ kPa) initiates a three-phase cascade: VCFM Phase 1 (Gamma Scattering, 40 Hz fails), VCFM Phase 2 (Alpha Detuning, 10Hz → 8Hz), and VCFM Phase 3 (Theta Dominance, low-pass filter). Each phase contributes cumulative signal loss feeding into the Clinical Consequence: Functional Rigidity and Signal Death.

Figure S2: Resonance Vulnerability Map (SymC Fault Lines)

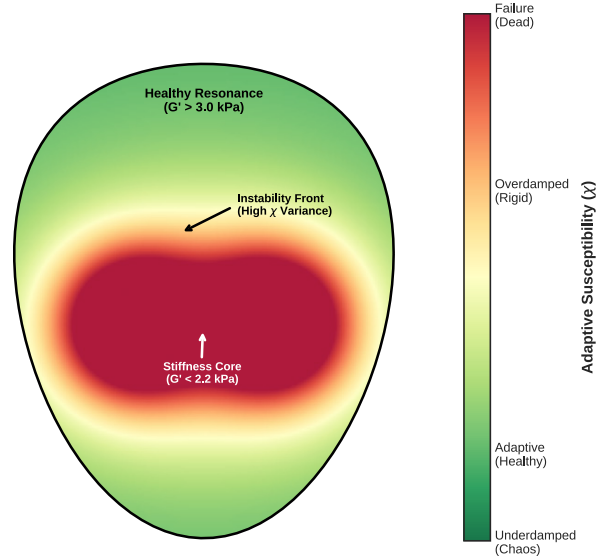


Figure 2: Resonance Vulnerability Map (SymC Fault Lines). Topographical visualization of brain vulnerability mapped onto the χ stability gradient. The outer region (green) represents Healthy Resonance ($G' > 3.0$ kPa) in the Adaptive Window. The Instability Front (yellow-orange) marks the zone of High χ Variance where fluctuations emerge. The Stiffness Core (red) indicates terminal failure ($G' < 2.2$ kPa). The colorbar maps Adaptive Susceptibility (χ) from Underdamped/Chaos through Adaptive/Healthy and Overdamped/Rigid to Failure/Dead.

Figure 3: Differential Temporal Cascade Map

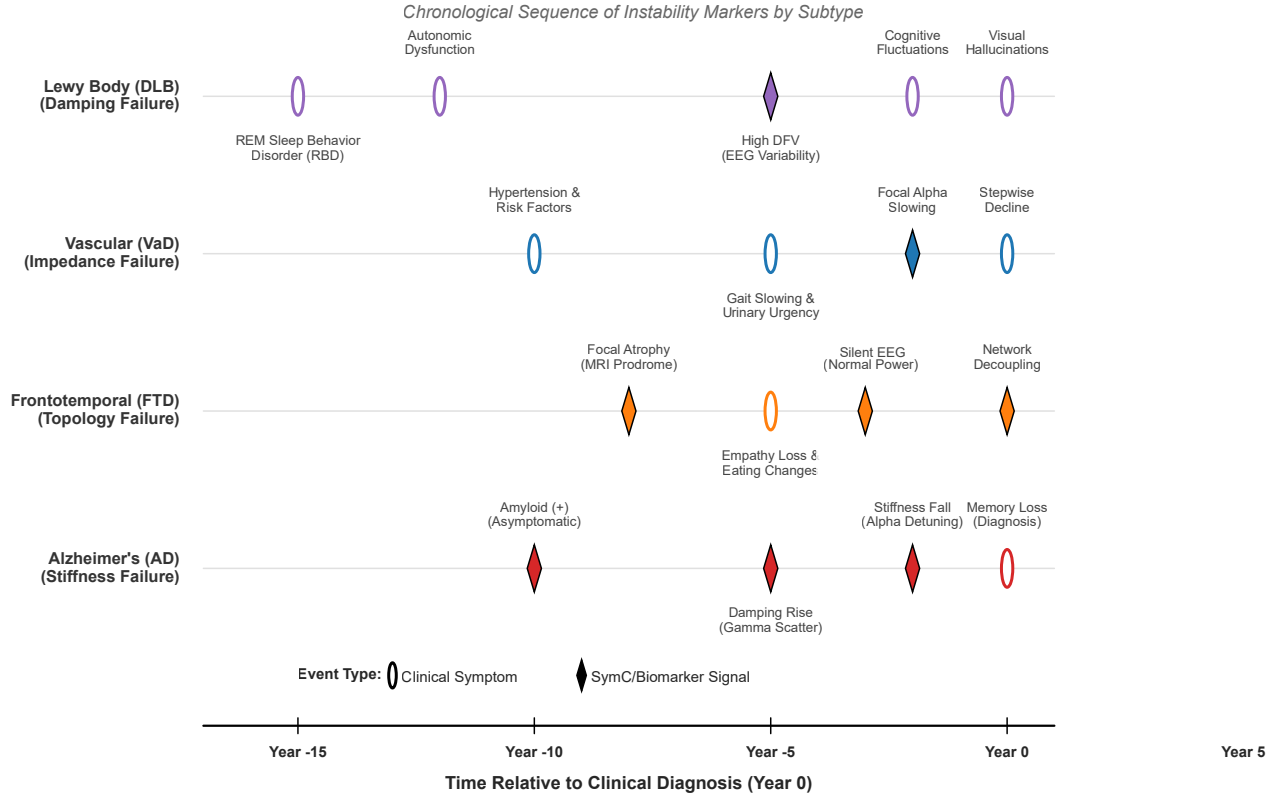


Figure 3: **Differential Temporal Cascade Map.** Chronological sequence of instability markers by subtype, aligned to clinical diagnosis (Year 0). Filled markers indicate SymC/biomarker signals; open markers indicate clinical symptoms. DLB shows the longest preclinical window (RBD at Year -15), while AD and FTD cluster markers closer to diagnosis. This visualization enables identification of subtype-specific “early warning” signatures.

S5: Mathematical Derivations

S5.1: Equivalence of ζ and ξ

For a damped harmonic oscillator:

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (2)$$

The damping ratio is:

$$\zeta = \frac{c}{2\sqrt{km}} = \frac{\gamma}{2\omega_0} \quad (3)$$

For a viscoelastic material under oscillatory shear:

$$G^* = G' + iG'' \quad (4)$$

The loss tangent is $\tan \delta = G''/G'$, and the viscoelastic damping ratio:

$$\xi = \frac{G''}{2G'} = \frac{\tan \delta}{2} \quad (5)$$

When neural tissue is modeled as a viscoelastic medium supporting wave propagation, the damping of oscillatory modes follows:

$$\zeta_{\text{wave}} = \frac{G''}{2G'} = \xi \quad (6)$$

Therefore $\chi \equiv \zeta \equiv \xi$ when appropriately defined in each domain.

S5.2: Stochastic Controller Dynamics (DLB)

The Ornstein-Uhlenbeck process for $\chi(t)$:

$$d\chi = -\frac{1}{\tau}(\chi - \chi_{\text{set}})dt + \sigma dW_t \quad (7)$$

Has stationary distribution:

$$p(\chi) = \sqrt{\frac{1}{\pi\sigma^2\tau}} \exp\left(-\frac{(\chi - \chi_{\text{set}})^2}{\sigma^2\tau}\right) \quad (8)$$

With variance:

$$\text{Var}[\chi] = \frac{\sigma^2\tau}{2} \quad (9)$$

In DLB, both $\sigma \uparrow$ (increased noise from neurotransmitter instability) and $\tau \uparrow$ (slower homeostatic correction from cholinergic deficit), yielding pathologically high variance.

S5.3: Impedance Mismatch Energy Reflection (VaD)

The reflection coefficient at a boundary:

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad (10)$$

For acoustic impedance $Z = \sqrt{\rho G'}$:

$$R = \frac{\sqrt{\rho_{\text{tissue}} G'_{\text{tissue}}} - \sqrt{\rho_{\text{vessel}} G'_{\text{vessel}}}}{\sqrt{\rho_{\text{tissue}} G'_{\text{tissue}}} + \sqrt{\rho_{\text{vessel}} G'_{\text{vessel}}}} \quad (11)$$

Assuming $\rho_{\text{tissue}} \approx \rho_{\text{vessel}}$:

$$R \approx \frac{\sqrt{G'_{\text{tissue}}} - \sqrt{G'_{\text{vessel}}}}{\sqrt{G'_{\text{tissue}}} + \sqrt{G'_{\text{vessel}}}} \quad (12)$$

In VaD: $G'_{\text{vessel}} \uparrow$ (arteriosclerosis) and $G'_{\text{tissue}} \downarrow$ (hypoperfusion damage), so $|R| \rightarrow 1$.

The reflected power fraction is $|R|^2$, meaning near-total energy reflection at advanced stages.

S5.4: Kuramoto Synchronization (FTD)

For N coupled oscillators with phases θ_j :

$$\dot{\theta}_j = \omega_j + \frac{K}{N} \sum_{k=1}^N \sin(\theta_k - \theta_j) \quad (13)$$

The order parameter:

$$r e^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j} \quad (14)$$

measures global synchronization. When frontal hub connectivity (K_{frontal}) degrades:

$$r_{\text{frontal}} \rightarrow 0 \quad (\text{decoherence}) \quad (15)$$

while posterior regions with preserved K maintain $r \approx 0.6$ – 0.8 .

S6: Information-Theoretic Derivation of the Adaptive Window

The claim that $\chi = 1$ represents the information-theoretic optimum derives from the Lindblad master equation governing open quantum systems, with direct classical analogues via fluctuation-dissipation relations.

S6.1: Lindblad Framework

For an open system with density matrix ρ , the Lindblad equation is:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (16)$$

where H is the system Hamiltonian (coherent evolution at rate ω) and L_k are Lindblad operators (dissipation at rate γ). The competition between coherent dynamics and dissipation is captured by:

$$\chi = \frac{\gamma}{2\omega} \quad (17)$$

S6.2: Information Efficiency at the Boundary

Information gain per unit entropy production is maximized when the system achieves fastest relaxation without oscillatory overshoot. For a second-order dissipative system:

$$\ddot{x} + \gamma \dot{x} + \omega^2 x = 0 \quad (18)$$

the discriminant $\Delta = \gamma^2 - 4\omega^2$ governs response character:

- $\Delta < 0$ ($\chi < 1$): Underdamped, oscillatory approach to equilibrium
- $\Delta = 0$ ($\chi = 1$): Critical damping, fastest non-oscillatory decay
- $\Delta > 0$ ($\chi > 1$): Overdamped, sluggish exponential decay

At $\chi = 1$, the system extracts maximum information from environmental perturbations (maintains observability) while dissipating minimum energy per bit processed (maximizes efficiency). This is the unique operating point where measurement-induced decoherence and environmental dissipation achieve optimal balance.

S6.3: Operational Window Under Realistic Constraints

Ideal systems would operate exactly at $\chi = 1$. Real systems face:

- Feedback delays (finite neural conduction velocity)
- Measurement noise (synaptic stochasticity)
- Bounded control authority (metabolic limits)
- Parameter uncertainty (biological variability)

These constraints shift the operational optimum slightly below the critical boundary to $\chi \approx 0.8$ – 0.95 , providing a stability margin against transient perturbations. This explains the empirical observation that healthy neural systems cluster in this range rather than precisely at $\chi = 1$.

S6.4: Neural Application

In neural tissue, the Lindblad operators correspond to:

- Synaptic dissipation (GABAergic inhibition $\rightarrow \gamma$)
- Coherent oscillation (glutamatergic drive $\rightarrow \omega$)

The viscoelastic equivalence $\xi = G''/(2G')$ maps tissue mechanics onto the same framework: G'' (loss modulus) corresponds to dissipation, G' (storage modulus) corresponds to elastic restoration. The identity $\chi \equiv \zeta \equiv \xi$ thus reflects a deep physical equivalence, not merely a mathematical coincidence.

Dementia represents departure from this optimum: information processing efficiency degrades as χ deviates from the adaptive window, manifesting clinically as reduced signal-to-noise ratio in cognitive operations before structural damage becomes apparent.

S7: Complete Reference List

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